Assessment Book — Chapter 2 – Shaping of the Earth’s Surface

Total marks: 15.0

Formative

Q1. [mcq] (1.0 marks) What is the rigid outer layer of the Earth that includes the crust and upper mantle called?

■ Asthenosphere

■ Lithosphere

■ Core

■ Mantle

Answer: Lithosphere

Concepts: Plate Tectonics

Q2. [short] (2.0 marks) Define a landform and give two examples.

Answer: A landform is a natural feature on the Earth's surface formed by processes such as weathering, erosion, deposition, and the movement of the Earth’s crust. Examples include mountains, valleys, plateaus, plains, deserts, and coastal features.

Rubric: Award 1 mark for the correct definition and 1 mark for providing two accurate examples.

Concepts: Plate Tectonics

Q3. [mcq] (1.0 marks) Which theory explains how large pieces of the Earth's crust move slowly over the molten mantle?

■ Theory of Continental Drift

■ Theory of Plate Tectonics

■ Seafloor Spreading Theory

■ Convection Theory

Answer: Theory of Plate Tectonics

Concepts: Plate Tectonics

Q4. [short] (2.0 marks) Distinguish between weathering and erosion regarding the movement of material.

Answer: Weathering is the process through which rocks break down into smaller pieces in place without involving the movement of the broken material. Erosion, on the other hand, is the process by which soil, rocks, and surface materials are worn away and carried from one place to another by natural agents.

Rubric: Award 1 mark for stating weathering does not involve movement of material, and 1 mark for stating erosion involves carrying material from one place to another.

Concepts: Weathering, Erosion

Summative

Q1. [mcq] (1.0 marks) Who is credited with giving the theory of plate tectonics?

■ Alfred Wegener

■ W.J. Morgan

■ Charles Darwin

■ Harry Hess

Answer: W.J. Morgan

Concepts: Plate Tectonics

Q2. [mcq] (1.0 marks) What feature is formed when continental plates collide?

■ Mid-ocean ridges

■ Fold mountains

■ Transform faults

■ Trenches

Answer: Fold mountains

Concepts: Plate Tectonics

Q3. [mcq] (1.0 marks) Which layer beneath the lithosphere is semi-molten and allows tectonic plates to move?

■ Core

■ Asthenosphere

■ Crust

■ Inner core

Answer: Asthenosphere

Concepts: Plate Tectonics

Q4. [short] (2.0 marks) Explain what causes tectonic plates to move.

Answer: The movement of tectonic plates is caused by convection currents in the mantle. Heat from the Earth's core causes molten material in the mantle to rise, while cooler material sinks, creating continuous currents that push and pull the tectonic plates.

Rubric: Award 1 mark for mentioning convection currents in the mantle and 1 mark for explaining the rising and sinking cycle driven by core heat.

Concepts: Plate Tectonics

Q5. [short] (3.0 marks) Describe the three main types of plate boundaries based on plate movement.

Answer: 1. Convergent boundary: two plates move towards each other, leading to collisions or subduction. 2. Divergent boundary: plates move away from each other, allowing magma to rise and form new crust. 3. Transform boundary: plates slide past each other without creating or destroying crust.

Rubric: Award 1 mark for each correctly described plate boundary type (Total 3 marks).

Concepts: Plate Tectonics

Q6. [long] (4.0 marks) Discuss the roles of weathering and erosion in the development of landforms, ensuring you correct the common misconception regarding rock movement.

Answer: Weathering and erosion play a vital role in the development of landforms by continuously breaking down and reshaping the Earth's surface. A common misconception is that weathering involves the movement of broken rock material; however, weathering strictly refers to rocks breaking down into smaller pieces in place without moving. Erosion follows or occurs alongside this by taking the worn-away soil, rocks, and surface materials and carrying them from one place to another using natural agents like water, wind, ice, or waves.

Rubric: Award marks for: explaining weathering (1 mark), explaining erosion (1 mark), explaining their combined role in shaping landforms (1 mark), and explicitly addressing and correcting the misconception about rock movement in weathering (1 mark).

Concepts: Weathering, Erosion

Q7. [short] (3.0 marks) Provide examples for each of the following: a fold mountain, a divergent boundary feature, and a transform boundary.

Answer: Fold mountain: The Himalaya. Divergent boundary feature: The Mid-Atlantic Ridge. Transform boundary: The San Andreas Fault in the United States.

Rubric: Award 1 mark for each correct example matching the specified boundary or feature type (Total 3 marks).

Concepts: Plate Tectonics